

Evaluation of Machine Learning Models for Real-Time Sign Recognition

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Abstract—Development of a real-time sign recognition system is a critical step in enabling communication between the speech impaired and normal individuals. This paper proposes a vision-based model for sign recognition using Machine Learning models. A local dataset is created by capturing the images under various light and background conditions by employing a webcam. AlexNet architecture and MATLAB are used to extract the features from captured images and to train several Machine Learning models for their performance evaluation. The images captured in real-time are processed to extract the features and then are directed to the classifier to predict the Sign class. The text to voice conversion module plays the speech based on the output of the classifier. An accuracy of 91.7 % is achieved with Support Vector Machine (SVM) learning model on the private dataset. Sign recognition with the proposed model is more reliable and cost-effective when compared to non-vision based approach.

Keywords—Machine Learning, MATLAB, Sign Language, Sign Recognition, Vision-based

I. INTRODUCTION

Communication plays a significant role in building one's life. Unfortunately, verbal communication for speech-impaired people becomes a barrier in articulating their ideas. Hence, they use sign language to communicate with others. But understanding sign language becomes challenging for ordinary people. Consequently, disabled people face difficulties in conveying things. An interpreter is required to translate the sign languages [1] for resolving this problem. Since there is no universal sign language, it becomes difficult to have an interpreter who knows all the sign languages. There are two automatic approaches mentioned in the literature to bridge this communication gap: vision-based and non-vision-based sign language translation systems.

Non-vision technique uses a wearable hand glove to recognize Signs. The system measures the changes in the resistance and hand orientation using flex sensors and accelerometers mounted on the glove. The output from these sensors is then processed by the microcontroller, to obtain the text and speech output based on the Sign. Vision-based systems [2] involves a human-machine interaction to make

communication efficient and feasible. A camera is employed to capture the signs, and the Signs are recognized using computer vision and Machine learning techniques [3]. In non-vision-based methods the circuit is more complex and might get damaged easily [4]. Also, it is expensive, and user may not feel comfortable in wearing the hand glove for longer durations.

A vision-based method is proposed in this paper, which is user friendly, cost effective and more reliable. The model performed effectively when tested under various illuminations, backgrounds and for different hand sizes and skin tones.

II. RELATED WORK

Bantupalli et al. [5] proposed a vision-based technique that uses iPhone 6 camera to capture the videos for the American Sign Language (ASL) dataset at 60fps. Signs are recorded with a full-sleeved shirt. Each video is broken down into several image frames. Later these image frames are augmented to expand the dataset. The user's hand Sign is recorded through a camera and the videos are preprocessed before feeding to the CNN inception model extraction of the temporal features from image frames. LSTM (Long short-term memory) classifies the segmented Signs into one of the Sign classes. The trained LSTM compares the results of SoftMax and global pool layers of the CNN. The system trained both the CNN and RNN techniques independently but failed to differentiate the skin tone. Also, the accuracy dropped due to the inclusion of faces because of variation in faces. The variation in clothing led to poor performance. Cross entropy cost function based on adaptive moments was used to minimize the loss. The model efficiency would have improved by isolating the hand Signs from the images using the ROI technique.

Athira et al. [6] proposed a single-handed dynamic sign identification with co-articulation elimination. Each recognition block is composed of sub-blocks to pre-process, extract the features, and classify the images. The system captures dynamic Signs using a web camera at a frame rate of 15fps. For a continuous Sign sequence, two successive signs are segregated with some fluctuation in gesticulation known as co-articulation, which is not considered for recognition.

Before the Sign is recognized, it is spotted and then the co-articulation area is eliminated by the acceleration feature. The pre-processing methods such as hand segmentation, face identification and elimination, associated component extraction, and noise reduction were examined in the segmentation process. The set of frames collected from various signers fixed the range of skin color in the YCbCr colorspace and was then used to segment the hand region. Convex Hull algorithm estimated the centroid and to identify the shape of the hand. Curve features and Zernike movements are extracted from keyframes to recognize the Sign using SVM classifier. Processing time is reduced in this model as it considers only keyframes in feature extraction. The drawback is that the Dataset was captured under optimum light conditions and uniform background. The performance was not evaluated for various illumination conditions.

Ahmed et al. [7] used a dataset that is composed of 3200 images, out of which 1200 images are from UNIP, and the remaining 2000 images are locally generated. The hand region was cropped and resized to 32x32 pixels using a python library named Open Cv while creating the dataset. The resized images were converted into gray-scale images and then fed into the model. The segmentation was done based on skin color threshold calculation. The features was extracted and then classified using CNN (Convolutional Neural Networks). The recognized hand sign is then translated into Bangla text using the translator API of Google translator. Capturing the images using Xbox Kinect or Leap motion cameras can improve the model's quality.

Muthu Mariappan et al. [8] proposed a vision-based real-time sign language recognition system. A camera is employed to capture the raw videos in a dynamic background and extracted image frames are resized. In the pre-processing, noise in the image frames is reduced by blurring the images. Later, these images are converted into HSV colorspace images and then processed to yield binary images. On binary images, morphological transformations are applied based on the shape of the image to obtain a clear foreground object. The Region of Interest (ROI) is then detected using a canny edge detection algorithm applied to the morphed images. Later required vector features are extracted from ROI for each image frame of the video. At the training time, extracted features are directed to the Fuzzy c-means algorithm (FCM) for partitioning the input data elements into a definite number of clusters. During testing, this algorithm compares the test file with the existing clusters and provides the id of the cluster centre that has the greatest degree of membership. The model achieved 75% accuracy for 40 ISL words. Although FCM is efficient, it takes more computational time than other models.

For continuous sign language recognition Anshul Mittal et al. [9] presented a hybrid model composed of leap motion sensor, neural network architecture and LSTM (Long Short-Term Memory). Firstly, 3D positions of fingertips are extracted from the sign with the help of Application Programming Interface (API). A total of twelve dynamic features are extracted and stored. Later the features in the sign sequence are normalized by the maximum value present in the sequence. A modified four gated LSTM cell with 2D Convolutional Neural Network (CNN) are fed with feature maps. LSTM contains ReLU (Rectified linear unit) activation function followed by a dense layer and one Softmax layer to classify multiple classes. A categorical cross entropy loss function is used to train and test the LSTM for a batch size of

256 over 1000 epochs. Average accuracies of 72.3% and 89.5% were obtained for the signed sentences and words respectively.

Saleh Ahmad Khan et al. [10] presented system in which initially a custom dataset is created and stored as a training and testing set for five signs. Flipped and Gaussian blurred images are used in training set of image augmentation. Images are preprocessed and trained using CNN. To check for accuracy, validation test was done for the dataset. A web camera connected to the GPIO pins of the Raspberry Pi translates the sign into speech. The classification was done with the help of ReLU and Softmax activation functions. Unique features are extracted using CNN. Each conversion layer of CNN is batch normalized and the Scikit label encoder normalizes the labels for each class. The text output is displayed on the 3.5-inch display connected to the Raspberry Pi 3B. System is portable and flexible. As a part of future work, GUI implementation can be done to this model.

Ohnmar Win et al. [11] proposed an automatic gesture-to-speech conversion system. The locally generated dataset was considered in which each sign represents a predefined word. Inbuilt laptop webcam captures the hand gesture made by the speech-impaired person. The user wore a red hand glove for convenience that helps in better detection. The RGB image is converted into grayscale then the color detection algorithm was used to subtract the color suppressed channel from the grayscale image to detect the red color object. The thresholding method converts this detected image into a binary image. COG (center of gravity) is estimated in the binary image. The de-noising procedure is applied to the binary image then the edges are detected using Convex Hull Algorithm. The text output is displayed using MATLAB software. This model eliminated the use of sophisticated technology and hence was cost-efficient.

Praveena et al. [12] proposed a system, where in the image frames from real time video are converted from RGB to grayscale using the Otsu's algorithm. Then, the SURF (Speeded up Robust feature) extracted features are applied to SVM (support vector machine) classifier. The recognized text output is displayed and then converted into voice output using a Microsoft-Sam speech synthesizer.

Rosemary et al. [13] presented a machine learning-based approach for sign language translation into text and voice output. A new dataset was created by assigning each sign with a word. Keras (Neural network library of python) and Tensor flow (open-source library) are used to build and train the model. OpenCV Lib of python and CNN are used for image processing. While CNN was used for segmentation, features were extracted based on skin color and images are classified using a softmax classifier. The text output is displayed on the laptop screen and the speech output is obtained from the voice module.

Nandan et al. [14] presented network training through the Kaggle dataset with 87,000 training and 87,500 images to test. The sample images for the dataset were captured in different light intensities and at various angles. A camera was employed to accumulate the series of images from the video. Later these images were preprocessed for background elimination, blob detection, noise reduction, grayscale transformation, brightness normalization, and scaling analysis. The preprocessed images were used for network training and were augmented during every epoch. After the augmentation

process, the Convolutional Neural Networks [8] recognizes the image patterns. The AlexNet architecture of CNN reduces over-fitting in the Dropout layer followed by every single Fully Connected layer. The model's accuracy was improved by this architecture, as it is composed of 7 hidden layers, 650K units, and 60M parameters.

Ankita Wadhawan et al. [15] presented a static sign conversion model with Deep learning methods. The fundamental step is data acquisition, which involves procuring RGB data from the static signs using a camera. The data is then processed, which includes noise removal using morphological techniques, image resizing, and normalization. Shuffled dataset is divided into validation and training set to train the model using Convolutional Neural Network (CNN). Tesla K80 Graphical Processing Unit is used for training the classifier model. Network is trained using algorithms with various optimizers for a maximum of 100 epochs with a categorical cross-entropy loss function. and has achieved an accuracy of 99.90%. However, the model could not recognize the dynamic signs.

In non-vision methods, many sensors must be mounted on the hand glove and there is a possibility of wear and tear leading to system failure. This paper proposes a camera-based Sign recognition system, which is user friendly.

III. PROPOSED METHOD

In this section, an image processing and Machine Learning based Sign recognition system is presented. The model mainly relies on Support Vector Machine as shown in Fig.1.

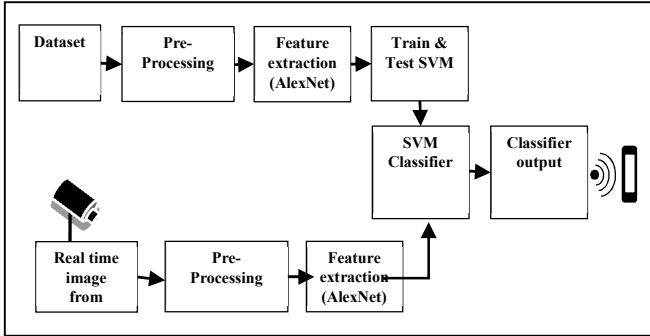


Fig. 1. Block diagram of sign language translator

SVM is a supervised machine learning algorithm that can be used to perform regression, classification, and anomaly detection. It chooses the data points that lie closest to the classification data. By determining the hyperplanes that distinguish the two classes, it will then proceed to produce the accuracy value. The obtained accuracy value depends on the value of kernel and the parameters that are used. The proposed model uses Non-Linear SVM with Cubic Kernel for classifying signs using multiclass classification. To perform multiclass classification, SVM divides the given Sign classes into multiple binary classifications.

The binary classification procedure involved in SVM can be described as follows:

Consider a training dataset of n points given as,

$$(x_1, y_1), \dots, (x_n, y_n)$$

where y_i can be 1 or -1 depending on the class to which x_i belongs. Each x_i is a real vector of dimension p . The maximum margin hyperplane is obtained by dividing the cluster of points x_i such that $y_i = 1$ and $y_i = -1$ lie on either side

of the plane. The distance between the maximum margin hyperplane and the nearest x_i point is maximized.

Equation of the hyperplane for the set of points x is given by,

$$w^T x - b = 0 \quad (1)$$

where w is normal vector to hyperplane, b is the bias.

From origin, the offset of the hyperplane along the normal vector is determined by the parameter $\frac{b}{\|w\|}$.

For the normalized dataset, the equations of hyperplane can be written as,

$$\begin{aligned} w^T x - b &= 1 \text{ for } y_i = 1 \text{ and} \\ w^T x - b &= -1 \text{ for } y_i = -1 \end{aligned} \quad (2)$$

The distance between two marginal hyperplanes is $\frac{2}{\|w\|}$ and it can be maximized by minimizing the parameter $\|w\|$.

The constraint i is added to prevent the falling of data points into the margin.

$$\begin{aligned} w^T x_i - b &\geq 1 \text{ for } y_i = 1 \text{ or} \\ w^T x_i - b &\leq -1 \text{ for } y_i = -1 \end{aligned} \quad (3)$$

The above equations can be generalized as

$$y_i (w^T x_i - b) \geq 1, \text{ for } 1 \leq i \leq n \quad (4)$$

Since the dataset is nonlinearly separable, the hinge loss function becomes helpful in obtaining the soft-margin.

$$\max (0, 1 - y_i (w^T x_i - b))$$

The above function is Zero when the constraint in the equation (4) is satisfied, which means the x_i lies on the appropriate side of the margin.

The purpose of optimization is to minimize

$$\left[\frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i (w^T x_i - b)) \right] + \lambda \|w\|^2 \quad (5)$$

Where λ is a parameter that determines the trade-off value between increasing margin size to ensure that the x_i point lies on the proper side of the margin.

The nonlinear transformation of data to a higher dimensional space is done by the kernel method to obtain the optimal linear solutions. The transformed data points are denoted as (x_i) and the kernel function is given as $\phi(x_i)$

$$k(x_i, x_j) = \phi(x_i) \cdot \phi(x_j) \quad (6)$$

In the transformed space, the real vector w satisfies,

$$w = \sum_{i=1}^n c_i y_i \phi(x_i) \quad (7)$$

where c_i are obtained by solving the problem of optimization. Maximize,

$$f(c_1, \dots, c_n) = \sum_{i=1}^n c_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i c_i (\phi(x_i) \cdot \phi(x_j)) y_j c_j \quad (8)$$

The pair wise inner product $(\phi(x_i) \cdot \phi(x_j))$ is replaced by the kernel function. Hence the above equation becomes,

$$f(c_1, \dots, c_n) = \sum_{i=1}^n c_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i c_i k(x_i, x_j) y_j c_j \quad (9)$$

subject to $\sum_{i=1}^n c_i y_i = 0$, and $0 \leq c_i \leq \frac{1}{2n\lambda}$ for all i .

In general, the kernel function will be

$$k(x_i, x_j) = (x_i \cdot x_j + 1)^d \quad (10)$$

where d is the degree of polynomial.

Therefore, the expression for cubic kernel used in the proposed work becomes

$$k(x_i, x_j) = (x_i \cdot x_j + 1)^3 \quad (11)$$

A camera is used to capture the images of different Signs under various backgrounds and illumination conditions, to create a dataset. The dataset is created with 10 Signs which are

numbered from 1 to 10. A set of abnormal images are acquired with two persons showing 10 different Signs with the variation in their arm extension and orientation. The dataset acquired consists of 1500 images, composing 150 images for each class. 80% of these images are chosen in random for training. The images are further resized to 227×227 and fed as input to the AlexNet for feature extraction. AlexNet architecture is used due to its high computational ability and its ability to use GPUs to train a large number of parameters. AlexNet extracts 1000 features using two normalization layers, three max-pooling layers, two fully connected layers, five convolutional layers, and one SoftMax layer. Convolutional layers are comprised of nonlinear activation function ReLU and convolutional filters. SoftMax is the output layer with 1000 channels where the extracted features are obtained. AlexNet contains 60 million parameters in total.

The 1000 features from AlexNet are then used to train the SVM model, and to predict and display the Sign class and the corresponding text in a MATLAB figure window. The model further plays the corresponding audio over loudspeakers.

IV. RESULTS AND DISCUSSION

A comparison of 5 Machine Learning algorithms, features, and other CNN models, with respect to accuracy was done and the results are listed in Table I-III. Further 13 parameters are calculated to evaluate which algorithm best suits for the dataset of Signs, and the results are provided in Table IV.

TABLE I. COMPARISON WITH OTHER CLASSIFIER MODELS

Classifier	SVM Cubic	Ensemble Subspace Discriminant	SVM Quadratic	Ensemble subspace KNN	KNN Fine
Accuracy	91.7%	90.5%	87.9%	85.3%	84.8%

TABLE II. COMPARISON WITH OTHER CNN MODELS

CNN Model	Alexnet	RESNET	VGG
Accuracy	91.7	87	85

TABLE III. COMPARISON WITH OTHER FEATURES

CNN Model	SIFT	SURF	HOG
Accuracy	86	85	84

The parameters calculated for each class are: False positive rate (FPR), True negative rate (TNR), True positive rate (TPR), False negative rate (FNR), accuracy, precision (P), recall (R), sensitivity (SE), specificity (SP), misclassification (MC), intersection over union (IoU), DICE similarity coefficient (DSC), pixel score (PxI) and Boundary F-1 score (BF score). It can be observed that True positive rates (TPR) of cubic SVM is comparatively high which resulted in more accurate prediction of signs. The obtained ROC curves are displayed in Fig.3 and Fig.4. From the tables and ROC curves, it is to be noted that amongst the models which have been compared, cubic SVM classifier model has got the highest accuracy and correspondingly excel in all the parameters.

Later the model is tested for real-time images acquired from Web camera. A sample random test image frame taken

from real time video (whose Ground truth is Sign 1), and the corresponding class result displayed in MATLAB figure Window is shown in Fig.2.



Fig. 2. Detected Sign Class 1

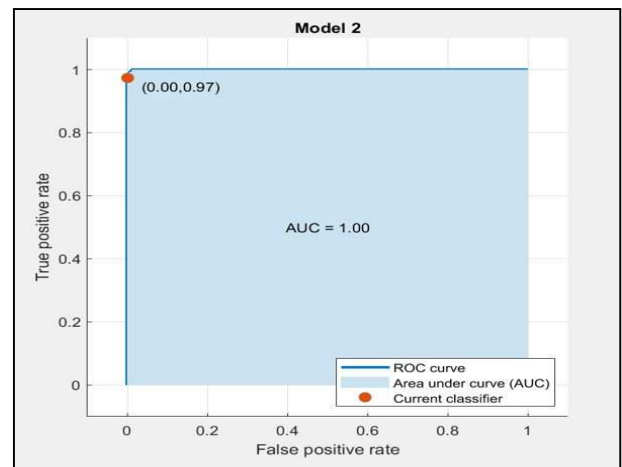


Fig. 3. SVM Cubic model

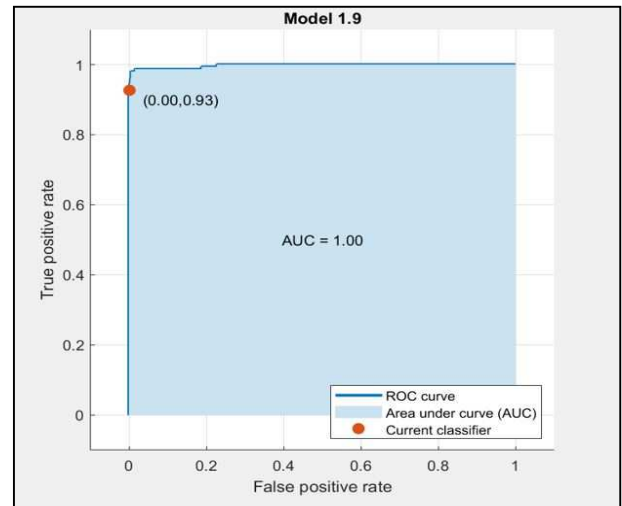


Fig. 4. SVM Quadratic model

A comparative study with other models is tabulated in Table 3. It can be concluded that the proposed model performed well in various light conditions and backgrounds. The model also predicted the signs correctly when tested for different skin tones, hand sizes, and orientations. Most of the systems mentioned in the Table V did not consider different

backgrounds and light conditions, skin tones together in the dataset.

TABLE IV. COMPARISON OF CLASSIFIER MODELS

SVM Cubic: Accuracy = 91.7%																	
Class	TP	TN	FP	FN	TPR	TNR	FPR	FNR	P	R	SE	SP	MC	IoU	DSC	Pxl	BF
1	146	1348	2	4	97%	99.85%	0.15%	3%	0.99	0.97	0.97	1.00	6.00	0.96	0.98	0.96	0.98
2	142	1342	8	8	95%	99.41%	0.59%	5%	0.95	0.95	0.95	0.99	16.00	0.90	0.95	0.90	0.95
3	132	1334	16	18	88%	98.81%	1.19%	12%	0.89	0.88	0.88	0.99	34.00	0.80	0.89	0.80	0.89
4	129	1334	16	21	86%	98.81%	1.19%	14%	0.89	0.86	0.86	0.99	37.00	0.78	0.87	0.78	0.87
5	140	1333	17	10	93%	98.74%	1.26%	7%	0.89	0.93	0.93	0.99	27.00	0.84	0.91	0.84	0.91
6	128	1336	14	22	85%	98.96%	1.04%	15%	0.90	0.85	0.85	0.99	36.00	0.78	0.88	0.78	0.88
7	149	1337	13	1	99%	99.04%	0.96%	1%	0.92	0.99	0.99	0.99	14.00	0.91	0.96	0.91	0.96
8	134	1338	12	16	89%	99.11%	0.89%	11%	0.92	0.89	0.89	0.99	28.00	0.83	0.91	0.83	0.91
9	133	1336	14	17	89%	98.96%	1.04%	11%	0.90	0.89	0.89	0.99	31.00	0.81	0.90	0.81	0.90
10	143	1338	12	7	95%	99.11%	0.89%	5%	0.92	0.95	0.95	0.99	19.00	0.88	0.94	0.88	0.94
SVM Quadratic: Accuracy = 87.9%																	
Class	TP	TN	FP	FN	TPR	TNR	FPR	FNR	P	R	SE	SP	MC	IoU	DSC	Pxl	BF
1	139	1348	2	11	93%	99.85%	0.15%	7%	0.99	0.93	0.93	1.00	13.00	0.91	0.96	0.91	0.96
2	137	1332	18	13	91%	98.67%	1.33%	9%	0.88	0.91	0.91	0.99	31.00	0.82	0.90	0.82	0.90
3	129	1327	23	21	86%	98.30%	1.70%	14%	0.85	0.86	0.86	0.98	44.00	0.75	0.85	0.75	0.85
4	123	1326	24	27	82%	98.22%	1.78%	18%	0.84	0.82	0.82	0.98	51.00	0.71	0.83	0.71	0.83
5	134	1328	22	16	89%	98.37%	1.63%	11%	0.86	0.89	0.89	0.98	38.00	0.78	0.88	0.78	0.88
6	125	1323	27	25	83%	98.00%	2.00%	17%	0.82	0.83	0.83	0.98	52.00	0.71	0.83	0.71	0.83
7	140	1337	13	10	93%	99.04%	0.96%	7%	0.92	0.93	0.93	0.99	23.00	0.86	0.92	0.86	0.92
8	126	1337	13	24	84%	99.04%	0.96%	16%	0.91	0.84	0.84	0.99	37.00	0.77	0.87	0.77	0.87
9	126	1329	21	24	84%	98.44%	1.56%	16%	0.86	0.84	0.84	0.98	45.00	0.74	0.85	0.74	0.85
10	139	1331	19	11	93%	98.59%	1.41%	7%	0.88	0.93	0.93	0.99	30.00	0.82	0.90	0.82	0.90

TABLE V. COMPARISON WITH THE STATE-OF-THE-ART LITERATURE

Model	Dataset	Feature Extraction	Classifier	Accuracy	F1 Score
Proposed model	Private	Alexnet	Cubic SVM	91.7%	0.88
Athira et al. [6]	Locally created	-	SVM	89%	-
Nandan et al. [14]	Kaggle	Alexnet	CNN	78.8%	0.78
Muthu Mariappan H et al. [8]	Locally created	-	Fuzzy C-means algorithm (FCM)	75%	-
Anshul Mittal et al. [9]	Locally created	Leap Motion sensor	Modified LSTM	89.5%	-

V. CONCLUSION

This paper presented a vision-based model to build an effective communication tool for speech-impaired people. The study evaluated performance of various machine learning techniques for classifying the Signs captured in real-time. It was observed that the cubic SVM classifier provides higher accuracy for AlexNet based features for the private dataset created. Although SVM is a binary classifier, the kernel trick being its major strength can convert non-linearly separable

data into high dimensional space for multiclass classification yielded high efficiency. The developed Machine Learning model is also reliable, cost-effective and yields better results than non-vision-based methods. In future, the model can be enhanced by increasing the number of sign classes in the dataset. Also, it can be advanced by creating an android application for sign recognition. The proposed work can also be extended to recognize the sign made by the user in a live

video conference. Exploitation of large datasets and detection of sign in complex backgrounds is the future work.

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